Briefing: what do we really know?

You're a junior analyst on the 2026-XJ desk. Separate fact from guess.

01 A

TASK 1 · SORT THE INTEL

Mark each line E (evidence), I (inference) or Q (question we can investigate).

An object is approaching Earth. ____

It will hit the school. ____

How fast is it moving? ____

Its exact path is uncertain. ____

It must be huge. ____

Where will it be in a week? ____

TASK 2 · PREDICT FROM DATA

A probe reads 10 m, 8 m, 6 m at t = 0, 1, 2 s. Predict x at 4 s. Name one reading that would challenge your model.

TASK 3 · WHAT DO WE NEED?

Name 3 measurements needed to predict where 2026-XJ will be next month.

TASK 4 · WHICH TOOLS?

Circle the Unit 1 tools that will help: position · velocity · acceleration · force · graphs · vectors. Pick one and say why.

ANALYST SIGN-OFF

Unit 1 asks: How fast, and where will it arrive? The first thing I want to measure is ____ because ____.

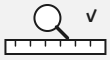
R1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 1 · R1 · SEI · SENTENCE FRAMES

Say it, then write it: fact or guess?

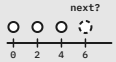
Language goal: I can **tell** evidence from a guess.



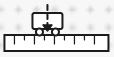
evidence
evidencia · evidência
what we saw or measured



inference
inferencia · inferência
a guess we still must test



prediction
predicción · previsão
what we think comes next



measure
medir · medir
find a number with a tool

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: evidence · inference · prediction · measure · test

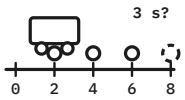
____ is something we saw or measured. An ____ is a guess that we still need to ____. To check a guess, we ____ with a tool. A ____ says what we think comes next.

2 · SAY IT TO A PARTNER, THEN WRITE IT



I know ____ is evidence because we ____ it.
____ is an inference because we still need to ____ .

3 · SOLVE WITH THE STEPS



A cart is at 2 m, 4 m, and 6 m at 0, 1, and 2 s.

STEP 1 · FIND THE PATTERN
each second it moves ____ m

STEP 2 · PREDICT
at 3 s: ____ + ____ = ____ m

STEP 3 · SAY IT
"My prediction is ____ m if ____."

4 · EXPLAIN IT · CHOOSE A FRAME

□ This is evidence, not a guess, because ____ .
□ My prediction could be wrong if ____ .
□ First we measured ____, then we predicted ____ .

I said it □ · I wrote it □

R1 1 2 3

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VOL. 1 · NO. 1 · R1 REPORTER: _____

The 2026-XJ Daily

Something is coming — report only the facts

BREAKING · WRITE THE HEADLINE (facts only!)

THE STORY

Observers have detected Object 2026-XJ approaching Earth. Write what is known, what is not known, and what scientists will measure next.

BY THE NUMBERS

What we know: ____
What we don't: ____
Measure next: ____
Physics tool: ____

LETTERS TO THE EDITOR

"Dear Editor: it's approaching, so it will definitely hit us." The editor replies:

SCIENCE DESK

A cart is at 2, 4, 6 m at 0, 1, 2 s. Predict 3 s. What assumption are you making?

CLASSIFIEDS

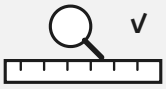
WANTED: one testable question about 2026-XJ. Must name a measurement. Write the ad:

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Picture dictionary

Language goal: I can tell a **fact** from a **guess**.



SEEN ✓
MEASURED ✓

evidence

evidencia · evidência

what we saw or measured

my language: _____

use it: _____



? → !

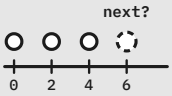
inference

inferencia · inferência

a guess we still must test

my language: _____

use it: _____



0 2 4 → 6

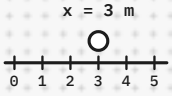
prediction

predicción · previsão

what we think comes next

my language: _____

use it: _____



x = 3 m

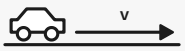
position x

posición · posição

where it is

my language: _____

use it: _____



+5 m/s →

velocity v

velocidad · velocidade

how fast + which way

my language: _____

use it: _____



gaps grow

v changes

acceleration a

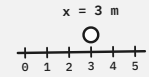
aceleración · aceleração

how velocity changes

my language: _____

use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE



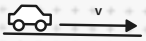
_____



_____



_____



_____

I can point and say all 6 words to a partner □

R1 (1) (2) (3)

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Settle the argument

01 I

DIRECTIONS · Shade the bubble of the person who is right. Give evidence, then fix the wrong claim.

ROUND 1

ANA
"The briefing proves 2026-XJ will hit Earth."

BEN
"It only says it's approaching. That is not proof of a hit."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
""Is it bad?" is a good science question."

LU
"Better: "Will its path reach Earth's position at the same time?""

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"Positions 2, 4, 6 m mean it will be at 8 m — guaranteed."

RAJ
"8 m is a prediction. It assumes the motion stays the same."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

R1 1 2 3

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Evidence, inference, or question?

01 N

DIRECTIONS · Read each card. Write its letter in the right column. Star the three trickiest.

A Observers detected an approaching object.	**B** It will destroy the school.	**C** Which direction is its velocity?	**D** Its path estimate is uncertain.
E It's coming at graduation, so it's aimed at us.	**F** How does its position change each day?	**G** It must be made of iron.	**H** The cart was at 2, 4, 6 m at 0, 1, 2 s.
I How big is it?	**J** Approaching means it is speeding up.	**K** Can a push change its path?	**L** The briefing says the story is fictional.

EVIDENCE	INFERENCE	QUESTION
stated or measured	a guess that needs testing	something we can investigate

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

R1 1 2 3

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Comic: evidence or just a guess?

01 CS

YOUR READER
A 4th grader who thinks every guess is a fact.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◯ **PENCIL** — An **E** next to the evidence in panel 1
- ☐ ◯ **PENCIL** — An **I** next to the inference in panel 2
- ☐ ◌ **BLUE** — A measurement with a number and unit in panel 3

1 · WE SEE IT *Show something a character can see or measure (example: wet ground).*

CAPTION _____

2 · WE GUESS *A character makes an inference about why it happened.*

CAPTION _____

3 · WE CHECK *Someone tests the guess with a tool or a new observation.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **evidence · inference · measure**

_____

_____

R1 1 2 3

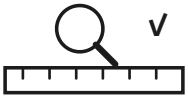
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Doodle Notes · observation

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Observation · observación (ES) · observação (PT)



TODAY'S CASE

Three photos show a cart at 2 m, 3 m and 4 m, one second apart.

01 · SKETCH AND LABEL

Draw the three measured positions as blue dots on a number line. Label each with its time.

02 · CHANGE THE PICTURE

If the pattern continues, where would the next dot go? Draw it with a dashed outline so a reader can tell it is a prediction.

Annotate your sketches. Circle the mark that best shows observation. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

A measured position differs from a prediction because …

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Frayer Model · observation

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Observation · observación (ES) · observação (PT)

A · It must have rained.

B · A photo shows wet pavement.

01 · DEFINITION

What does observation mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

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